Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	*(a)*			Conversion of kg to newtons (1) Application of upward forces = downward forces (1) Force = 62.2 [N] (1)	1	 1 1		 3	 1	
	(b)	(i)		Calculating components of velocity (11.6 hor, 13.8 vert) Calculating time of flight (2.81 s) or **alternative** e.g. $R = \frac{v^2 \sin 2\theta}{g}$ (2 marks) Horizontal range = 32.5 [m] **ecf** on time of flight (1) No, kick would **not** be successful – conclusion (1)			4	4	3	
		(ii)	I.	The (sum of) the products of mass times distance squared (1) from axis (of rotation of the ball) (1)	2			2		
			II.	Substitution into $J = I\omega$ (1) Angular momentum = 0.23 [kg m^2 s^{-1}](1)	1	 1		 2	 1	
			III.	Substitution into KE = $\frac{1}{2}I\omega^2$ (1) Rotational KE = 6.3 J [**ecf** on ω](1) Realising speed at greatest height = 18 $\cos$ 50^o (1) Total KE = 36(.4) [J] (1)	1	 1 1 1		 4	 4	
	(c)	(i)		$\frac{1}{2}\rho A v^2 C_D$ used (1) Stating $\rho v^2 C_D$ remain constant (can be implied) (1) Drag force [increased] - factor of 5.04 for end over end (1) Or drag force factor 5.04 [less] for spin type kick	1	 1 1		 3	 1	
		(ii)		The values of C_D, speed and A stay the same for different altitude (1) Factor would not change (1)		1	 1	 2		
				Question 8 total	**6**	**9**	**5**	**20**	**10**	**0**